Experiment 03_3 — CamemBERT Deep Learning Fine-Tuning

Goal: Evaluate transfer learning on French product texts using pre-trained `camembert-base` and compare it with the TF-IDF + SVM baseline.

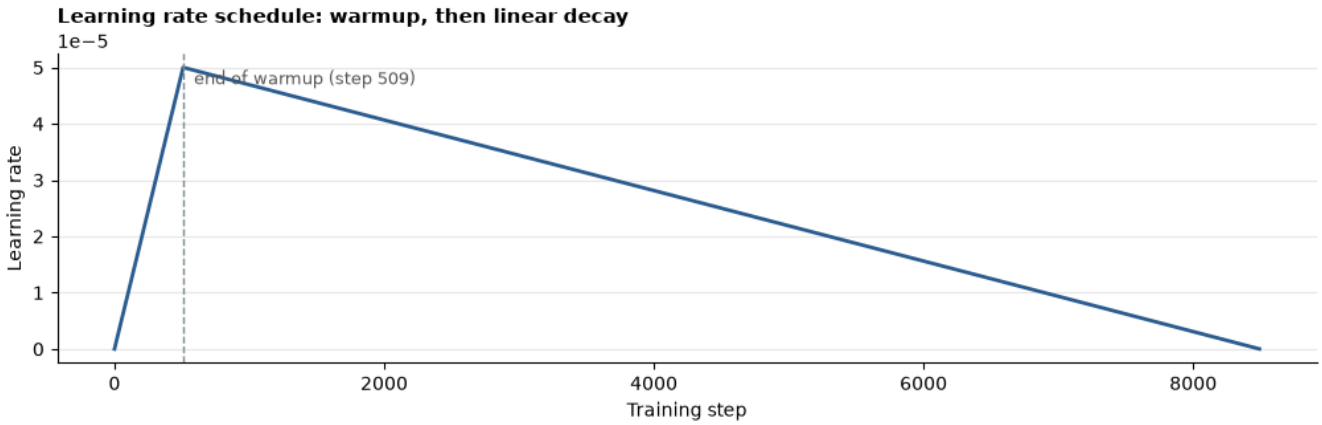
1. **Architecture & Setup**: Pre-trained CamemBERT transformer (110M parameters) fine-tuned on title and description text.
2. **Learning Dynamics**: Convergence inspection via 500-step training and validation loss tracking with linear warmup/decay schedule.
3. **Per-Category Performance**: Head-to-head comparison of per-class F1 gains against the optimised LinearSVC model.
4. **Error Analysis**: Systematic evaluation of top confused class pairs between traditional and deep learning approaches.

1. Training setup

Setting	Value
Model	CamemBERT base (110 million parameters)
Input	Title and description, truncated to 128 tokens
Epochs	2
Batch size	16
Optimiser	AdamW
Learning rate	0.00005, linear warmup (6% of steps) then linear decay
Weight decay (L2 regularisation)	0.01
Dropout	0.1
Gradient clipping	1.0

These are standard values for fine-tuning BERT-type models. They were not tuned with cross-validation, because one training run takes about 2 hours.

- Weight decay and dropout are regularisation: they reduce overfitting.
- Gradient clipping limits very large updates and keeps training stable.



Analysis

- Warmup (first 509 steps): the learning rate increases from 0 to 0.00005. The classification layer starts with random weights, so large updates at the start could damage the pre-trained weights.
- Linear decay: the learning rate then decreases to 0, so the final updates are small.

2. Training

The code below fine-tunes CamemBERT on the 67932 training products. One run takes about 2 hours on an Apple M1 Pro GPU.

With `RUN_TRAINING = False`, the notebook skips training and uses the saved results from the last run. Set it to `True` to train again.

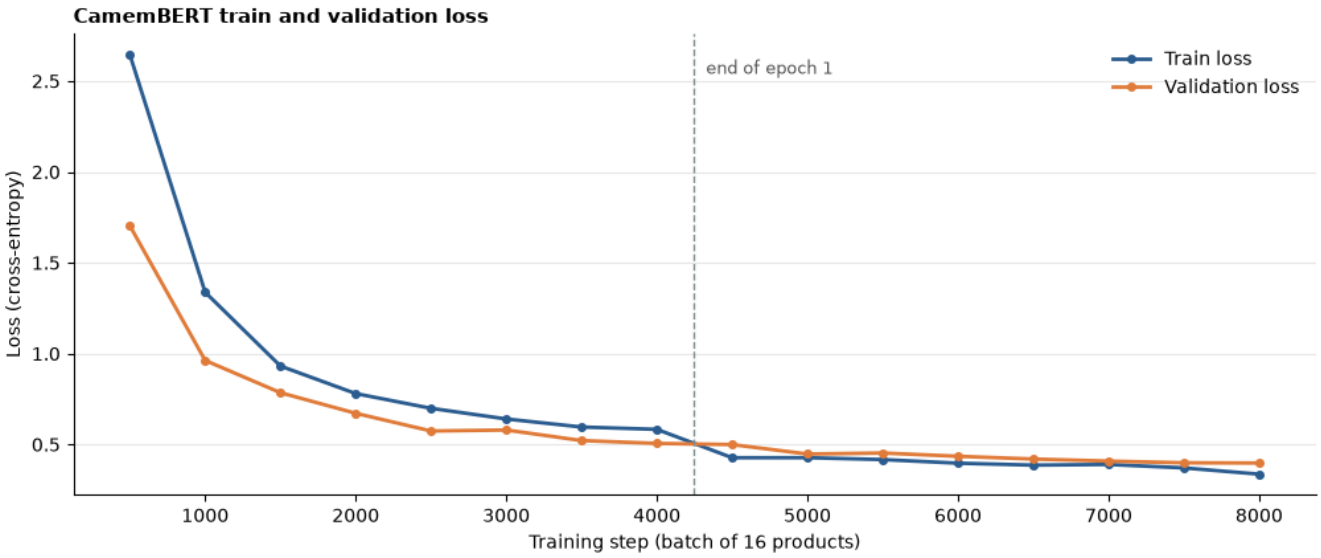
```
Training skipped: using the saved results of the last run.
```

3. Training curve

Every 500 steps, we measure:

- Train loss: the average cross-entropy loss over the last 500 training batches.
- Validation loss: the cross-entropy loss on a fixed sample of 2000 validation products.

If the validation loss stops decreasing while the train loss keeps decreasing, the model is overfitting.



Epoch	Validation loss	Validation weighted F1	Validation macro F1
1	0.5010	0.8527	0.8314
2	0.4269	0.8765	0.8594

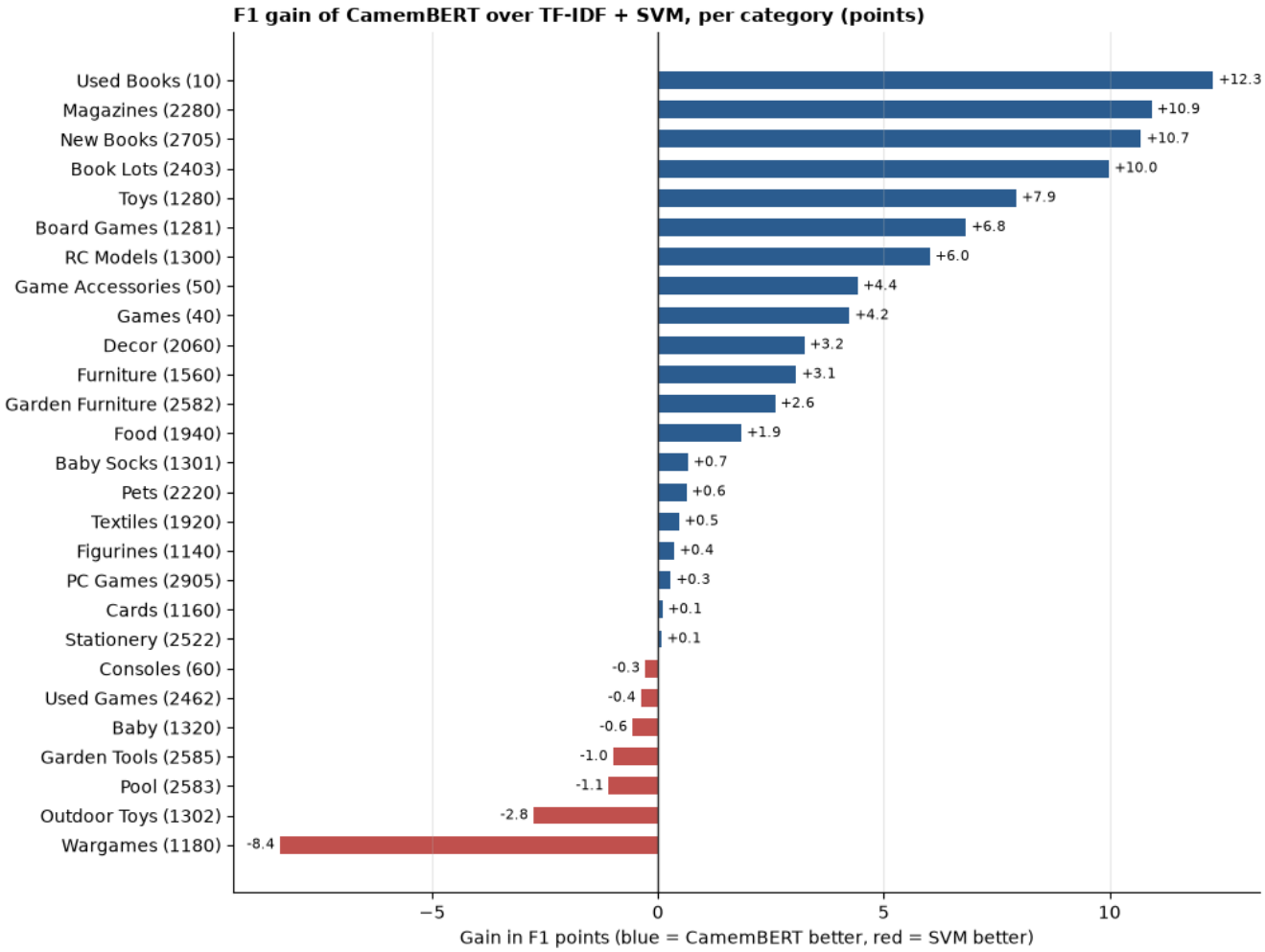
Analysis

- Both losses fall quickly in the first 1000 steps, then more slowly.
- Validation loss keeps going down until the end: 0.501 after epoch 1, 0.427 after epoch 2. So the model is not overfitting.
- In epoch 2, train loss drops a little below validation loss, because the model sees each training product for the second time. The gap is small (0.34 against 0.40 at the last measure). A third epoch could start to overfit, so 2 epochs is a reasonable stop.
- Validation weighted F1: 0.8527 after epoch 1, 0.8765 after epoch 2.
- The train loss point at step 4500 is rescaled: it was logged as an average over 500 steps but covered only the 254 steps since epoch 2 started.

4. CamemBERT vs TF-IDF + SVM

Both are scored on the same 16984 validation products. The SVM is the optimised model from notebook 03_2.

Model	Weighted F1	Macro F1
TF-IDF + SVM	0.8430	0.8326
CamemBERT	0.8765	0.8594



Analysis

- CamemBERT beats TF-IDF + SVM by 3.4 points of weighted F1 (0.8765 against 0.8430) and 2.7 points of macro F1.
- The largest gains are on the book categories: Used Books +12.3 points, Magazines +10.9, New Books +10.7, Book Lots +10.0. These were the weakest categories for the SVM.
- Toys, Board Games and RC Models also gain 6 to 8 points.
- The SVM is still better on a few categories, mainly Wargames (-8.4) and Outdoor Toys (-2.8). These depend on rare, specific words (*warhammer*, *heroclix*, *trampoline*) that the SVM uses directly.

5. Most confused pairs

The 8 category pairs CamemBERT mixes up most often, with the number of errors for each model.

```
Total errors: CamemBERT 2082, SVM 2656
```

True category	Predicted as	CamemBERT errors	SVM errors
Used Books (10)	New Books (2705)	97	78
Toys (1280)	Board Games (1281)	63	51
New Books (2705)	Used Books (10)	58	90
Book Lots (2403)	Used Books (10)	50	77
Book Lots (2403)	Magazines (2280)	46	95
Board Games (1281)	Toys (1280)	44	91
Toys (1280)	Outdoor Toys (1302)	43	27
Toys (1280)	Figurines (1140)	40	37

Analysis

- CamemBERT makes 2082 errors against 2656 for the SVM (22% fewer).
- The hardest pairs are still the same: Used Books and New Books, Toys and Board Games, Book Lots and Used Books.
- CamemBERT halves the confusion between Book Lots and Magazines (46 errors against 95) and between Board Games and Toys (44 against 91).
- It confuses Used Books with New Books more often than the SVM (97 against 78). Both are books with similar titles, and only the condition differs.

Conclusion

- CamemBERT is the best single text model: 0.8765 weighted F1 on validation.
- It helps most where the SVM was weakest, the book categories, because it reads words in context instead of counting them.
- The cost is training time: about 2 hours on a laptop GPU, against a few seconds for the SVM.